Check for updates

Received: 30 January 2023 | Revised: 31 January 2024 | Accepted: 2 February 2024

DOI: 10.1111/ajps.12860

AJPS AMERICAN JOURNAL of POLITICAL SCIENCE

ARTICLE

Political accountability under moral hazard

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Abstract
Viewing the relationship between politicians and voters as a principal–agent interaction afflicted by moral hazard, we examine how political careers are shaped by the incentives that voters provide incumbents to work in the public interest. When moral hazard binds, the optimal way for voters to hold politicians accountable is to provide re-election incentives that evolve dynamically over their careers in office. Under these incentives, first-term politicians are among the most electorally vulnerable and the hardest-working; politician effort rises with electoral vulnerability; electoral security increases following good performance and decreases following bad performance; and both effort and electoral vulnerability tend to decline with tenure. In extensions, we study limited voter commitment, voluntary retirement from politics, and adverse selection.

A longstanding view in the political accountability literature that originated with the seminal works of Barro (1973) and Ferejohn (1986) is that the relationship between voters and politicians is a principal–agent relationship that is afflicted by moral hazard. For instance, if the economy stagnates under an incumbent officeholder—if inflation rages or if unemployment soars—voters cannot easily tell whether it is because of mismanagement by the politician, or factors outside her control. Given this uncertainty, what is the best way for voters to hold politicians accountable if their relationship is a potentially long-term one, and the only lever of control voters have is the decision of whether to re-elect or replace the incumbent officeholder?

Prior work on political accountability (surveyed by Ashworth, 2012, and Duggan & Martinelli, 2017) has focused mainly on settings in which the politician and voter are short-lived, or in which the politician is term-limited; or, it imposes the ad hoc restriction that voters condition their vote choice only on outcomes observed during the incumbent's present term in office. From a positive standpoint, there may be good reasons to impose these restrictions. But in this paper we are concerned with the normative question of what re-election rules are optimal for voters when they are constrained by nothing other than their own credibility.

The model that we consider is very similar to that of Ferejohn (1986). It features a representative voter and a set of politicians, one of whom is in office each period of time. In every period, the politician in office privately observes a shock that takes a binary value (good/high or bad/low) and is drawn independently each period from an identical distribution. When the shock is good, the sitting politician has control over the outcome and voter welfare is determined by the politician's action in office. When the shock is bad, the politician has no control over the outcome, and voter utility is guaranteed to be low no matter what the politician does.

After observing the shock, the politician chooses her action, which represents the extent to which she serves the public interest at the expense of her private interest. (Thus, a key feature of our model, as in any accountability setting, is the lack of perfect congruence between the voter and the politician.) For simplicity, and following the broader accountability literature, we refer to the politician's action as *effort*, though it could represent the choice of any policy that favors the voter more while favoring the politician less.

Politician effort, together with the shock, determines performance. Under the bad shock, the politician cannot deliver for the voter, and so there is no reason not to act in her private interest, that is, put in no effort. Under the good shock, performance

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increases linearly with effort. Crucially, the voter cannot tell whether a bad outcome was due to a bad shock or due to low effort. Finally, the voter chooses whether to retain the politician or replace her with a new one from the pool.[1]

In our model, a re-election rule that considers only the politician's performance in her present term in office is not always optimal. Instead, when the misalignment of interests between the voter and politician is high (as measured by the marginal cost of effort), the optimal re-election rule has the voter conditioning re-election on a broader history of the politician's performance in office. In particular, under the optimal rule, politicians have careers, with voters extracting more service out of early-term politicians via the reward of future job security. A politician's electoral security evolves stochastically and tends to grow with time in office. With certainty, some politician eventually becomes fully entrenched in office.

We also examine three variants of our model. First, we suppose that the voter will credibly renegotiate away from any continuation play that delivers him a low payoff. Similar dynamics arise in this case as in the baseline model, except that no politician ever becomes fully entrenched. Second, we look at the case in which politicians receive stochastic outside job offers while in office. We show that voters may prefer to have the politician take these offers, thus outsourcing past effort's reward to the private sector. Third, we add adverse selection to the model by assuming that politicians have types (good or bad) and type is complementary to effort. We show that a novel dynamic trade-off arises between sanctioning bad performance and selecting good types. This trade-off is absent from existing two-period models. Together, these results reveal the versatility and tractability of the canonical accountability model when we move beyond the traditional focus on stationary incentives.

Existing work on political accountability does not reveal much in terms of long-run career dynamics, as it focuses on stationary re-election rules or short time horizons. For example, Ashworth (2005) studies a three-period career concerns model in which politicians allocate effort across different activities, focusing on how career stage determines this allocation; and Mattozzi and Merlo (2008) study a two-period model in which politicians decide whether to enter/remain in politics or work in the private sector. Yet, prior literature has documented the fact that politicians do have evolving careers. Hibbing (1991), for example, shows that while incumbents do very well in general, first-term members of Congress are the most electorally vulnerable and re-election rates among those seeking a new term is generally increasing in tenure. A variety of factors shape the political lifecycle, including the returns to seniority in the party and the legislature (McKelvey & Riezman, 1992; Muthoo & Shepsle, 2014), increases in productivity and skill that come with job experience (Padró i Miquel & Snyder, 2006), changes in individual, constituent, and ruling party ideology (Ansolabehere & Snyder, 2002; Jacobson, 2015), and the creation of a loyal voting coalition among constituents, which potentially grows more stable over time (Fenno, 1978). Our model shows that if voters were to implement the normative benchmark, then the unobserved occurrence of idiosyncratic shocks could confound the link between political career dynamics and factors such as seniority, learning-by-doing, political consolidation, and so on.

Models in which politicians are unrestricted in the number of terms they can serve typically focus on stationary equilibria following the early work of Barro (1973), Ferejohn (1986), Banks and Sundaram (1990, 1993), and others. An exception is Schwabe (2009), who allows politicians to serve an unrestricted number of terms but restricts attention to a class of equilibria that are suboptimal for the voter. In recent work, Anesi and Buisseret (2020) and Kartik and Van Weelden (2019) also allow for an unrestricted number of terms, studying models with both adverse selection and moral hazard. Anesi and Buisseret (2020), however, restrict attention to the case of minimal discounting while Kartik and Van Weelden (2019), who look at settings both with and without term limits, focus on a class of Markovian equilibria in the latter.[2] Our paper differs from this work by focusing on long-run career dynamics under optimal accountability when stationary equilibria are suboptimal.

Our paper also relates to prior work on repeated moral hazard. It is most closely related to work studying principal-optimal equilibria in settings in which the principal is unable to finely adjust the agent's compensation. This includes the work on delegation by Lipnowski and Ramos (2020), Li et al. (2017), and Guo and Hörner (2020), as well as other studies in political economy, including those on indirect control and war by Padró i Miquel and Yared (2012) and Yared (2010) and a recent paper by Foarta and Sugaya (2021), who look at principal-optimal equilibria in an intervention game. In these contributions, as in our work, the standard recursive toolbox developed by Spear and

[1] For tractability, we depart from the Ferejohn (1986) model in two ways. First, we assume the shock is binary rather than continuous. Second, we assume effort is bounded above, and the politician has a constant marginal cost of effort, whereas Ferejohn (1986) assumes effort is unbounded with increasing marginal cost. We explain at the end of the "Career Dynamics" section how the main qualitative results of our analysis carry over if we assume effort is unbounded and/or has increasing marginal cost (while retaining the assumption of binary shocks).

[2] In a setting in which the agents/politicians have heterogeneous expertise in assessing risky decisions, Aghion and Jackson (2016) show tenure contracts are approximately optimal for a patient principal/voter, and explore the value of term limits when the voter cannot commit.

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Srivastava (1987) and Abreu et al. (1990) facilitates the analysis of how the future terms of a relationship can substitute for monetary incentives. These techniques have been applied much more broadly in a variety of settings—for example, in related work by Fong and Li (2017), Clementi and Hopenhayn (2006), DeMarzo and Fishman (2007), Thomas and Worrall (1990), and Atkeson and Lucas (1992), which all share with our work the feature that utility is imperfectly transferable due to limited liability or risk aversion.

A distinguishing aspect of our work is that our repeated principal–agent interaction has all three of the following features: the principal (i) cannot fine-tune the agent's monetary rewards (e.g., options, bonuses, commissions), (ii) can choose to replace or retain the agent at the end of each interaction (the voter's so-called "blunt tool"), and (iii) incurs no cost to replace the agent. In addition to being a realistic feature of the political agency relationship, these features together furnish additional structure by endogenizing the principal's outside option. This allows us to further enrich the political agency model, which we demonstrate by studying settings in which the voter can renegotiate with the politician, the politician can voluntarily retire from politics, and adverse selection is also present in the agency relationship.

BASELINE MODEL

The players in the model are a representative voter and an infinite collection of politicians. Time is discrete, with an infinite horizon and indexed by $t \in \{0, 1, 2, ...\}$. Each period starts with a politician in office. One of the politicians begins in office at date 0. In each period, a political state variable θ_t is drawn from $\{0, 1\}$ independently across periods and privately seen by the officeholding politician. In state $\theta_t = 1$, the officeholder has the ability to serve the voter's interests, while in state $\theta = 0$ she does not. Hereafter, we refer to $\theta_t = 1$ as the good/high shock and $\theta_t = 0$ as the bad/low shock. We assume that in each period the good/high shock has probability $\mu \in (0, 1)$ and the bad/low shock has probability $1 - \mu$.

After seeing the shock, the politician in office chooses how much to serve the public interest, $a_t \in [0, 1]$. High levels of a_t means that she serves the public interest at the expense of her private interest, while low levels mean the opposite. Following the broader accountability literature, we refer to a_t simply as "effort," but it should be understood to mean effort at serving the public interest.

Finally in each period t, the voter observes only her own payoff, the product $y_t = \theta_t a_t$, and publicly decides whether to retain the politician or remove her from office. Slightly abusing terminology, we will refer to y_t as the *outcome* of period t and will often drop the time subscript when the period is clear. We denote by ρ_t the probability with which the politician is re-elected. If the voter removes the politician, she is replaced by someone from the pool and never re-enters office in the future.[3]

In each period t, the politician in office accrues a flow payoff $1 - ca_t$, where $c > 0$ is the marginal cost of effort, which measures the degree of conflict of interest between politicians and voters.[4] All politicians not in office get 0. All individuals discount the future using a common discount factor, $\delta \in (0, 1)$. In periods $t > 0$, we will use the term *incumbent* to refer to the politician in office in the previous period; for period $t = 0$, the incumbent is the initial politician in office. We express continuation values in terms of average flow payoffs. We look at perfect public equilibria of this model that are optimal from the voter's perspective.[5]

Observe that as far as on-path behavior is concerned, voter-optimal equilibrium is equivalent to the strategically simpler setting in which the voter can commit to a strategy. That is, even though our solution concept is voter-optimal equilibrium (taking voter incentives into account), studying this simpler setting with commitment yields the same predictions about the dynamics of political accountability.

To see this equivalence, considering any strategy profile σ that respects politician incentives and makes the voter indifferent whenever mixing, we can modify it to an alternative strategy profile with the same on-path behavior that additionally respects the voter's incentives. Although there are typically many such equilibria, one example is the strategy profile that agrees with σ at all of its on-path histories, and has every politician shirking and being immediately replaced at all of its off-path histories. At any history at which the voter is expected to fire or re-elect the politician, doing so will be incentive compatible (IC) because behaving differently will lead to zero effort in the future.[6] So the only relevant voter incentive constraint is that he be indifferent between firing and re-electing whenever expected to mix. We will ignore

[3] The latter is without loss for voter utility. For any equilibrium in which the politician is re-elected after some delay, some weakly better equilibrium exists in which delayed re-election never occurs in that equilibrium.

[4] We have thus fixed the office benefit to 1 in addition to fixing the maximum effort bound to also be 1, but this is without loss of generality. A setting in which the politician's office benefit is $B > 0$, marginal cost of effort is $C > 0$, and maximum politician effort in a period is $\bar{a} > 0$ is equivalent to ours if we divide politician utility by B, voter utility by $\bar{a}$, and politician effort by $\bar{a}$—leading to a specification of our model with $c = \bar{a}C/B$. Comparative statics of various equilibrium quantities with respect to B, C, and $\bar{a}$ can be taken after accounting for these rescalings. For example, the voter's highest equilibrium value is increasing in $\bar{a}$, increasing in B, and decreasing in C.

[5] See Definitions 5.2 and 5.3 in Fudenberg and Tirole (1991) for the relevant definition of a perfect public equilibrium, which we refine by focusing on voter-optimal equilibrium.

[6] An alternative construction is one in which every politician behaves in the same way as the politician who was "supposed to be" in office. This equilibrium strikes us as more plausible, but is more cumbersome to describe formally.

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this constraint in our analysis for our baseline model—it will be satisfied by the voter-optimal re-election rule described in Propositions 1 and 2.[7] We thus let an IC policy refer to a strategy profile under which no politician has an incentive to deviate from her prescribed conditional effort choice at any history.

Next, observe that a special kind of IC policy is without loss of optimality from the voter's perspective. Specifically, each politician uses a pure strategy; each public history h has some effort $a[h]$ such that the politician in office exerts effort zero given a bad shock and effort $a[h]$ given a good shock; and continuation play is identical for all outcomes $y \geq a[h]$ and identical again for all outcomes $y < a[h]$. We will refer to $a[h]$ as the equilibrium-prescribed level of effort at history h; and we will refer to any outcome $y \geq a[h]$ as a *success* and to any outcome $y < a[h]$ as a *failure*. For such a policy, it is apparent that a politician's incentives will be satisfied as long as she always willingly chooses the equilibrium-prescribed effort over an effort of zero, following a good shock. Given this form, from now on we will abuse notation and interpret a politician's choice at public history h to be her choice at history h conditional on a good shock in that period.

To see that the above-described policies are without loss of optimality, first note that a politician can always use a pure strategy when the voter is committed: If she were mixing over multiple pure strategies—all of which would then be best responses—choosing a voter's favorite (or near-favorite) among them would offer a weak improvement. Moreover, working is never optimal for a politician under the bad shock, since it leads to an inferior payoff in the stage game with no effect on the public outcome of the dynamic game. So when the shock is bad, the politician in office chooses not to exert any effort. Now, let $a[h]$ denote the effort that the politician chooses following a good shock, and define success and failure as in the previous paragraph. Consider a modification to the voter's strategy in which he responds to every failure as if the outcome were 0 and every success as if it were $a[h]$. This policy generates the same voter payoff as the original policy because it generates the same path of play. So, all that remains is to verify that it is IC.

To see that it is IC, note that under the modified voter behavior, a politician always prefers to choose 0 over $a \in (0, a[h])$ and $a[h]$ over $a > a[h]$. Therefore, the modified policy is IC as long as the politician weakly prefers $a[h]$ to zero following a high shock. As these two choices produce outcomes that are treated in the same way as under the original policy (which was assumed to be IC), the modified policy is IC as well. Hence, the given form of equilibrium is without loss in terms of voter payoffs. Throughout the paper, we focus only on such policies.

The voter's first-best payoff is μ, which obtains if and only if every politician in office exerts maximum effort in every high-shock period.

Before building on the above observations to study voter-optimal policies, we use them to study the benchmark class of stationary equilibria. The following proposition reports the main features of optimal stationary equilibria and shows that they are characterized by a cutoff retention rule.

Proposition 0. *In a voter-optimal stationary equilibrium, for a politician in office in period t, we have the following:*

1. *The politician chooses effort $a_t = \theta_t a^*$, where $a^* = \min\{1, \delta/c\}$.*
2. *The politician is fired if she fails ($y_t < a^*$) and retained if she succeeds ($y_t \geq a^*$).*

The expected equilibrium payoff to the voter is $\mu a^* = \mu \min\{1, \delta/c\}$. Thus, a stationary equilibrium exists that achieves the voter's first-best payoff if and only if $\delta \geq c$.

The proofs of all formal results in this paper appear in the Online Appendix.

CAREER DYNAMICS

In the previous section, we showed that if $\delta \geq c$, a stationary equilibrium achieves the voter's first-best payoff, so in this case, some stationary equilibrium is obviously voter-optimal. We now look at the case in which $\delta < c$ and study political career dynamics under voter-optimal equilibrium play. Before proceeding, we note that we have not assumed that $c < 1$; therefore, the analysis of this case also concerns the case of high δ (arbitrarily close to 1) if the marginal cost of effort is $c \geq 1$.

We take a recursive approach à la Spear and Srivastava (1987). More specifically, we think of the voter as solving a dynamic programming problem whose relevant state variable is the in-office politician's continuation value from the current period onward. We start by noting that the politician's continuation value u starting from any period must lie in the interval $[1-\delta, 1]$. She can guarantee herself a continuation payoff of $1-\delta$ by shirking indefinitely, and can attain a payoff of at most 1 by accruing the benefits of office in every period without ever exerting any effort. The voter can generate any continuation payoff for the politician in the interval $[1-\delta, 1]$ by randomizing between two extremes—immediate replacement after the current term and the offer of unconditional full job security.

[7] In our extension with adverse selection, firing will have an (endogenous) cost, so we will augment the model with a public randomization device to allow the voter to optimally use a pure strategy.

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Recall (as noted in the previous section) all successes are treated equally, and all failures treated equally, at a given history. Thus, the optimal reelection rule takes the politician's continuation value $u \in [1-\delta, 1]$ at the start of the period to two possible continuation values at the start of the subsequent period, the value v_s associated with the politician succeeding to deliver the voter's equilibrium prescribed payoff in the current period, and the value v_f associated with the politician failing to do so.

To characterize the law of motion of politician continuation values, two observations of the voter's optimal value function are useful. First, the function is concave: Variation in the politician's continuation value is costly to the voter. Second, the function is decreasing: Firing the current politician weakly benefits the voter since the newly elected politician could be offered the same continuation equilibrium as the in-office one. This property tells us that a newly elected politician begins her career at continuation value $1-\delta$.

The law of motion of politician continuation values then satisfies two constraints. The first is the voter's promise-keeping (PK) constraint, which says that a politician's starting continuation value is the value that the voter must give her:

$$u = \mu[(1-\delta)(1-ca) + \delta v_s] + (1-\mu)[(1-\delta)1 + \delta v_f]. \tag{PK}$$

The other constraint is the politician's IC constraint, which says that the politician must have an incentive to put in the equilibrium prescribed effort a when the shock is good:

$$(1-\delta)(1-ca) + \delta v_s \geq (1-\delta)1 + \delta v_f. \tag{IC}$$

For the voter to be optimizing, this constraint in fact holds with equality—because concavity of the optimal value function means superfluous variation of the politician's continuation value is costly to the voter. Thus, the (PK) constraint and the (IC) constraint holding with equality together give us two equations in two unknowns, namely, v_f and v_s. Solving these equations gives us the exact form for the law of motion of politician continuation values:

$$v_u(a, y) := \begin{cases} v_s = \frac{1}{\delta}[u - (1-\delta)(1-ca)] & \text{if } y \geq a, \\ v_f = \frac{1}{\delta}[u - (1-\delta)] & \text{if } y < a. \end{cases}$$

We now specify the politician's equilibrium prescribed effort a as a function of her continuation value u at the start of the period; denote this a_u. We show this effort is as high as the politician's incentives (and the promised continuation value of u) will allow: It is the maximum possible effort level of 1 if the continuation value is low enough, and otherwise makes the politician indifferent when success is rewarded with perfect future job security.[8] That is, since v_s cannot exceed 1, when u is close enough to 1, we must have the equilibrium prescribed effort be $a_u = (1-u)/[(1-\delta)c]$.

Finally, because the politician's continuation value conditional on being retained cannot be below $1-\delta$, if u is low enough, then following failure by the politician we must have the voter mix between retaining the politician at continuation value $1-\delta$ from the next period on, and firing her to give her a payoff of 0. To guarantee that the politician's continuation value following failure is indeed v_f, this mixing probability must be $v_f/(1-\delta)$ for small enough values of u. Moreover, that the value function is decreasing lets us show the voter should only fire at this minimum feasible rate.

We now have our main result for the baseline model, which summarizes the observations above about voter-optimal accountability.

Proposition 1. *Suppose $\delta < c$. In a voter-optimal equilibrium, every first-term politician takes office with continuation value $u = 1-\delta$; and for any politician who starts any period t with continuation value $u \in [1-\delta, 1]$, we have:*

1. *The politician chooses $a_t = \theta_t a_u$, where $a_u = \min\left\{1, \frac{1-u}{(1-\delta)c}\right\}$.*
2. *Given the outcome y in period t, if $v_u(a_u, y) \geq 1-\delta$, then the politician is retained with probability 1 at continuation value $v_u(a_u, y)$ starting from the next period. If $v_u(a_u, y) < 1-\delta$, she is retained with probability $v_u(a_u, y)/(1-\delta)$ at continuation value $1-\delta$ from the next period.*

Moreover, every voter-optimal equilibrium generates the same distribution over the possible paths of play.

The behavior described in Proposition 1 has the politician working as hard as can be made IC (given the politician's current continuation value)—and hence being made indifferent between working and shirking—and the voter replacing the politician as little as possible.[9]

[8] This feature is the only feature of Proposition 1, or its proof, that relies on our assumptions of bounded effort with linear costs. In the general case (outlined under the "General costs" heading at the end of this section), a closed form is inaccessible, but one can use concavity of the voter's optimal to show that a_u is weakly decreasing in u.

[9] Although stationary equilibria become optimal at $\delta = c < 1$, the model's predictions are not discontinuous at this point. For example, the probability of a given politician becoming completely entrenched converges to zero as $\delta < c$ converges to c.

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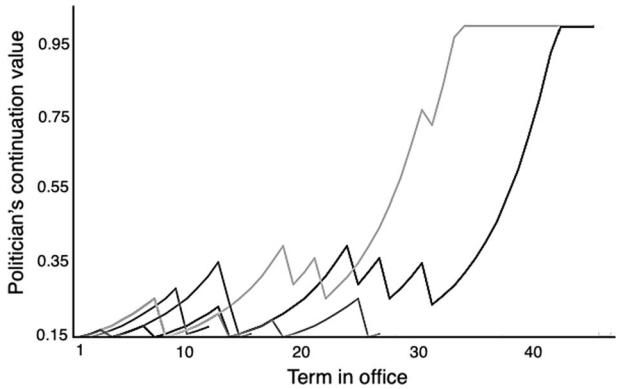

FIGURE 1 Simulated continuation value paths. *Note*: The figure shows examples of simulated paths of politician continuation values for the following parameter values: $\mu = 0.8$, $c = 0.9$, and $\delta = 0.85$. The important thresholds are: $1 - \delta = 0.15$, $1 - \delta^2 = 0.2775$, and $1 - (1 - \delta)c = 0.865$.

As a politician's continuation value climbs with a stream of successes, it is possible for her to earn full job security on the path of play. Whenever her continuation value is above $1 - (1 - \delta)c$, being successful in the current term results in complete entrenchment. The corollary below shows that with a long enough run of successes, the politician's continuation value can cross this threshold.

Corollary 1. *Suppose $\delta < c$, and let $k \in \mathbb{N}$ be smallest such that $\frac{\delta}{1-\delta^k} \leq c$. Then, any politician who has k consecutive successes will be completely entrenched.*

We now make a series of observations. First, because the sequence of cost cutoffs in the corollary converges to δ, for any value of $c > \delta$, some politician will eventually achieve enough consecutive successes to become entrenched in office.

Second, the corollary puts an upper bound on the number of consecutive successes needed for the politician to become completely entrenched. But, in fact, a politician may become entrenched with fewer consecutive successes if she has a sequence of successes following a failure that resulted in her being re-elected with a continuation value larger than $1 - \delta$.

Third, to give a political interpretation to the conditions stated in the corollary, note that since the conditions do not rely on μ, we can interpret δ as reflecting both the frequency of elections and how voters and politicians trade off the future against the present. Given this interpretation, the corollary implies that the minimum time to entrenchment is shorter if elections are less frequent, if politics is more short-termist, and if the conflict of interest c between the voter and politicians is greater.

Figure 1 makes our observations concrete by depicting some simulated paths of the politician's continuation value for a numerical example. It shows that some politicians have longer careers than others. Some become completely entrenched more quickly; others take longer; and still others are removed from office before they achieve full job security. In the early stages of a politician's career, failure is more prone to result in the politician being replaced or their careers being reset (i.e., the continuation value falling to $1 - \delta$). Failure later on in a politician's career is less likely to result in removal or a reset. A politician is more likely to serve another term the more successful she has been in the past, but even politicians who start their careers with a good run may have short careers if they are unlucky enough to experience a sequence of bad shocks. Therefore, while job security tends to grow over a politician's career in office, it need not grow monotonically. Similarly, a politician's effort tends to decline over her career in office, but also need not decline monotonically.[10]

We emphasize that our observations about the path of play are not merely features of a special voter-optimal equilibrium characterized in Proposition 1. These observations are strengthened by the uniqueness result stated in the proposition: All voter-optimal equilibria are equivalent in terms of the paths of play that they generate. We prove this in Online Appendix Section A.2 (pp. 3–8).

Finally, in Online Appendix Section B (pp. 23–24) we numerically compute the voter's optimal equilibrium payoff for different parameter values, and compare it to his payoff in the best stationary equilibrium and his first-best payoff. The computations show that there are significant welfare gains from removing the restriction to stationary equilibria.

General costs. The qualitative structure of the voter's preferred equilibrium transports to a more general version of our model with weakly increasing marginal effort cost, and effort allowed to be bounded or unbounded. This specification includes the case of unbounded effort with strictly convex costs, exactly as in the Ferejohn (1986) formulation. In this more general specification, each history still has only one relevant IC constraint for the politician in office, which continues to hold with equality in the voter's preferred equilibrium; and the politician's continuation value following the realization of the voter's utility within the period is still one of two quantities that we can continue to call the values after "success" and "failure." The politician's continuation value will still rise with success and fall with failure. The voter's decision to

[10] Whereas average effort is highest for newly elected politicians and converges to its lowest level as tenure in office converges to infinity, even average effort may not decline monotonically over time, because time is discrete in our model. For example, when $\delta/(1 - \delta^2) < c < \delta/(1 - \delta)$ and μ is very small, one can verify that the average third-term politician exerts higher effort than the average second-term politician.

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retain or replace the politician will continue to reflect the same dependence on the politician's continuation value, and the politician's equilibrium effort will still be decreasing in her continuation value. All of these observations can be proven by following the exact same proof, mutatis mutandis, as that of Proposition 1.

Stationary equilibria remain suboptimal whenever moral hazard binds. If effort is bounded, then a threshold discount factor exists such that stationary equilibria will achieve the first-best voter payoff above this threshold, and stationary equilibria will be strictly suboptimal below it; and if effort is unbounded, stationary equilibria will always be strictly suboptimal.

While the general insights carry over, we opt to focus on the simpler setting in which effort is bounded and its marginal cost is constant because this case further enables an exact closed-form characterization of politician effort at each continuation value, and hence of the voter-optimal equilibrium. As the following three sections of the paper show, having such a closed-form solution facilitates various detailed extensions that let us further explore the role of dynamics in political careers.

The other aspect of our model that differs from Ferejohn (1986) is the shock distribution: We assume binary shocks whereas Ferejohn (1986) has continuous shocks. This feature of our model is an important tractability assumption.

RENEGOTIATION-PROOF EQUILIBRIA

The results of the previous section show how the voter incentivizes work from an incumbent politician (in the early and middle parts of the politician's career) through the delayed reward of full job security. But when a politician becomes completely entrenched, the voter can expect to derive no subsequent value from her. Can the voter commit to tolerating this low value? Though Proposition 1 describes a perfect equilibrium of the game (and so, interpreted literally, requires no commitment power), a valid concern is that voters might want to somehow coordinate on a new equilibrium when facing such a dismal future.

We now limit the voter's ability to commit to tolerating these low future payoffs by imposing a threshold, $\underline{\pi}$, such that his equilibrium payoff cannot fall below $\underline{\pi}$ at any history. This approach is reminiscent of the concept of renegotiation-proofness introduced by Pearce (1987). In Proposition 2, we consider exogenous thresholds below which the voter's payoff cannot fall at any equilibrium history. One can think of the *extent* of the voter's commitment power as being measured by the magnitude of $\underline{\pi}$: The higher $\underline{\pi}$ is, the less ability the voter has to tolerate unfavorable outcomes ex post. After documenting play that will hold for any such threshold $\underline{\pi}$, we then turn (in Corollary 2) to endogenizing $\underline{\pi}$. In line with Pearce (1987), we focus on the highest threshold that can possibly be sustained, essentially allowing the voter to renegotiate away from the path whenever such renegotiation is itself credible.[11]

Given an exogenous limit to the voter's ability to commit, the next proposition shows that the sole effect of this limit on voter-optimal play is to cap the politician's achievable continuation value.

Proposition 2. *Suppose $\delta < c$, and fix a payoff lower bound $\underline{\pi} > 0$. Some $\hat{u} \in [1-\delta, 1)$ exists that is decreasing in $\underline{\pi}$ such that, in any equilibrium that is optimal for the voter subject to the constraint that the voter's continuation payoff after every history is at least $\underline{\pi}$: (i) The continuation value at any history of any politician in office lies in $[1-\delta, \hat{u}]$; (ii) every politician who enters office starts with continuation value $1-\delta$; and (iii) if a politician starts a period t with continuation value u, we have the following:*

1. *The politician chooses $a_t = \theta_t a_{u,\hat{u}}$, where $a_{u,\hat{u}} = \min\{1, \frac{(1-\delta)(1-u)+\delta(\hat{u}-u)}{(1-\delta)c}\}$.*
2. *Given the outcome y in period t, if $v_u(a_{u,\hat{u}}, y) \geq 1-\delta$, then the politician is retained with probability 1 at continuation value $v_u(a_{u,\hat{u}}, y)$ starting from the next period. If $v_u(a_{u,\hat{u}}, y) < 1-\delta$, then she is retained with probability $v_u(a_{u,\hat{u}}, y)/(1-\delta)$ at continuation value $1-\delta$ from the next period.*

An important implication of the proposition is that the optimal way to "protect" the voter from suffering low payoffs is to directly preclude entrenchment—that is, to place a permanent upper bound on the degree of job security a politician can enjoy, whatever her history is in office. For any level of voter security $\underline{\pi}$, the proposition says that an "entrenchment limit" $\hat{u}$ exists that could assure the voter a value of $\underline{\pi}$ from any history, and moreover that a more stringent voter security level necessitates a more stringent limit to politician entrenchment. If a politician ever achieves the highest possible continuation value of $\hat{u}$, then she chooses a low effort of $(1-\hat{u})/c$ in that period, maintains the same continuation value if she succeeds, and has her continuation value fall to the strictly lower value of $[\hat{u}-(1-\delta)]/\delta$ if she fails. Apart from this modification, the dynamics of a politician's career are qualitatively similar to those of the baseline model.[12]

The main justification for a lower bound $\underline{\pi}$ on continuation payoffs is that the voter might renegotiate

[11] While our approach is inspired by Pearce (1987), one key difference is that Pearce (1987) chiefly focuses on strongly symmetric equilibria of a symmetric game, while we apply analogous reasoning to a principal–agent setting assuming that the principal can unilaterally initiate renegotiation.

[12] Another implication that follows immediately from the proposition is that, for the purposes of limiting politician entrenchment, it is suboptimal to impose a fixed term limit of any length.

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away from an equilibrium when continuation play gives him a low payoff, say $\pi_L < \underset{\sim}{\pi}$. However, this renegotiation will be credible only if the proposed new equilibrium is not vulnerable to the same sort of renegotiation. Internal consistency requires that, at any possible future history, the new equilibrium continuation payoff that the voter faces is itself at least π_L because, otherwise, the voter would similarly wish to renegotiate. This consistency requirement can be taken a step further by asking for a sort of external consistency: If an equilibrium exists that guarantees a continuation payoff of $\pi_H > \pi_L$ at every possible future history, then the equilibrium that guarantees only π_L not be credible. If a history from which the voter's continuation is near π_L might again occur, would the voter not renegotiate away from this new equilibrium? And would the politicians not anticipate this? Following Pearce (1987), only one level of worst-case continuation payoff to the voter, $\underset{\sim}{\pi}$, satisfies both consistency properties: the highest feasible lower bound $\underset{\sim}{\pi}$. Intuitively, a higher bound is unattainable, whereas a lower bound would not be immune to renegotiation.

One may wonder whether choosing $\underset{\sim}{\pi}$ to be the endogenous payoff lower bound ultimately selects (or at least allows for) stationary equilibria. The following corollary shows that, at least for some parameter values, this is not the case.

Corollary 2. *If $1/(2-\mu) < \delta < c$, then some $\underset{\sim}{\pi}$ exists such that some equilibrium gives the voter a payoff of at least $\underset{\sim}{\pi}$ after every history, while every stationary equilibrium gives the voter a payoff strictly below $\underset{\sim}{\pi}$. In particular, an equilibrium of the form described in Proposition 2 exists such that the voter is strictly better off after every history than under any stationary equilibrium.*

The corollary shows that allowing the voter to renegotiate away from bad equilibria (i.e., limiting voter commitment) can actually *preclude* stationary play. Following Pearce's justification that renegotiation itself must be credible (which selects $\underset{\sim}{\pi}$ from Proposition 2 to be as large as possible), Corollary 2 shows that equilibria featuring richer dynamics can be "more credible" for voters than stationary equilibria. Note that the assumption in the proposition that $\delta < c$ simply rules out the case in which the voter can achieve his first-best payoff. The condition $1/(2-\mu) < \delta$ says that, despite binding moral hazard, the future is still sufficiently valuable.

Put differently, the conclusion of Corollary 2 leads us to question the conventional view that stationary equilibria describe the natural outcome when voters cannot commit in the political accountability model. The result shows that this interpretation is misleading. After all, what exactly is the voter's "inability to commit"? We have argued (following Pearce) that it is the common understanding that the voter may, in some contingencies, desire to renegotiate the promises he has made to the politician. But from where does this desire to renegotiate arise? Presumably, it is activated by the voter facing a pessimistic enough future. Lack of commitment, then, is the voter's inability or unwillingness to tolerate low future payoffs, which is exactly what the Pearce bound formalizes. Lack of commitment, importantly, is *not* the voter's inability or unwillingness to condition on past history.

VOLUNTARY RETIREMENT

Many politicians voluntarily retire from office and find lucrative positions outside of politics, such as in the private sector, as lobbyists, advisers, industry executives, and board members of major corporations and organizations. For example, in recent work, Egerod (2021) finds that when indicators of job prospects outside of politics improve, sitting politicians become more likely to retire from politics, while Hall and Van Houweling (1995) note that politicians that are more electorally insecure are more likely to retire from politics (see, also, Diermeier et al., 2005; Groseclose & Krehbiel, 1994; Keane & Merlo, 2010).

Suppose we enrich the baseline model to allow for voluntary retirement by assuming that in each period, a politician receives with some probability the opportunity to leave her career in politics and work elsewhere. In what follows, we study how this possibility affects the politician's optimal incentives and career dynamics.

Specifically, consider a modified model in which the in-office politician randomly receives outside offers—giving her utility $w > 0$—and can, thus, decide to voluntarily retire from politics. The rate of arrival of these offers depends on whether or not the politician is in office. The in-office politician receives an offer with probability $p \in (0,1)$, while an out-of-office politician receives an offer with probability $\underline{p} \in [0, p)$. Without loss of generality, we normalize $\underline{p}$ to equal zero—or, equivalently, interpret politician payoffs as being payoffs net of her continuation value after being replaced. In addition, we restrict attention to the interesting case of $w \in [1-\delta, 1]$. If $w < 1-\delta$, every offer is turned down, and the ability to work outside of politics has no effect on political career dynamics. If $w > 1$, every offer is accepted, and while the arrival of these offers does affect a politician's continuation value, her career dynamics are the same as in a version of the baseline model with a lower discount factor, δ, and a higher benefit from holding office (rather than the normalized benefit of 1 in our baseline model). Finally, it will be convenient to give the players

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access to a public randomization device, also known as "sunspots."[13]

Each period t proceeds as follows. First, the sunspot is revealed. The politician in office then either privately receives an outside offer of w with probability p, or no offer with probability $1-p$. If the politician has received an outside offer, she decides whether to accept or reject it. If she accepts the offer, a new politician arrives and potentially receives an outside offer to accept or reject. After some politician in office either does not receive or does not accept an outside offer, she receives her office benefit of 1, observes the state θ_t, and makes an effort choice a_t at a private cost of ca_t. Then, the outcome $y_t = \theta_t a_t$ is publicly observed. Finally, the voter decides whether to re-elect the incumbent or to replace her.[14]

Proposition 3. *Suppose $\delta < c$ and $1-\delta \le w \le 1$. In a voter-optimal equilibrium, every first-term politician starts in office with continuation value $(1-p)(1-\delta) + pw$, and some $v^* \in \{w, 1\}$ and $u^* \in [w, 1]$ exist such that the politician's continuation value at the start of every period lies in the interval $[(1-p)(1-\delta) + pw, v^*]$, and for any politician who starts any period t with continuation value $\tilde{u}$, we have the following:*

1. *If $\tilde{u} \le w$, then she accepts an outside offer if it arrives, and her continuation value after the outside-offer stage (if she does not accept one) is $u = \frac{\tilde{u}-pw}{1-p}$. If $\tilde{u} > u^*$, then she does not accept the outside offer even if one arrives, and her continuation value after the outside-offer stage is $u = w$. If $\tilde{u} \in (w, u^*]$, then with probability $\frac{u^*-\tilde{u}}{u^*-w}$ (determined by the sunspot), she accepts an outside offer if it arrives, and her continuation value after the outside-offer stage (if she does not accept one) is $u = w$; with probability $\frac{\tilde{u}-w}{u^*-w}$, she does not accept the outside offer even if it arrives, and her continuation value after the outside-offer stage is $u = u^*$.*
2. *Given the politician's continuation value u after the outside-offer stage, she chooses $a_t = \theta_t a_{u,v^*}$, where $a_{u,v^*} := \min\{1, \frac{(1-\delta)+\delta v^*-u}{(1-\delta)c}\}$.*
3. *Given the outcome y in period t, if $v_u(a_{u,v^*}, y) \ge (1-p)(1-\delta) + pw$, then the politician is retained with probability 1 at continuation value $v_u(a_{u,v^*}, y)$ starting from the next period. If $v_u(a_{u,v^*}, y) < (1-p)(1-\delta) + pw$, she is retained with probability $v_u(a_{u,v^*}, y)/[(1-p)(1-\delta) + pw]$ at continuation value $(1-p)(1-\delta) + pw$ from the next period.*

The proposition characterizes a voter-optimal equilibrium up to two details. The first unspecified feature is whether the highest admitted continuation value for a politician is w or 1. Second, if it is the latter, one must specify the length of the interval (w, u^*) in which the politician mixes between taking and leaving the outside offer when it is available.

When the highest possible continuation value for the politician is 1, effort decisions are the same as in the baseline model, conditional on the politician's continuation value. The politician and the voter use the public randomization device only to coordinate on continuation play in the interval above w in which the politician mixes between taking and leaving the outside offer. In this case, note that the voter has the discrete benefit of having the politician leave office by taking the outside offer and starting afresh with a new politician to whom he owes nothing.

When the politician's highest possible continuation value is w, effort is reined in (as it is in Proposition 2 to limit entrenchment) to keep the politician's value from exceeding w. This way, the voter benefits from the politician taking the outside offer whenever it arrives, again due to the benefit of getting to start with a new politician who enters office at the lowest possible continuation value on the equilibrium path. Here, because the politician's continuation value is always below w, the public randomization device is never used.

Finally, note that when $v^* = w$, politicians never become entrenched, accepting every outside offer that arrives on the path of play. They will vacuously do so when $w = 1$, but a more detailed calculation is required in the general case. The next corollary establishes that a sufficient condition for this property is that $w < 1$ is high enough.

Corollary 3. *Suppose $\delta < c$. If $w < 1$ is sufficiently high, then a voter-optimal equilibrium exists such that every politician has a hazard rate of at least p of leaving office in any given period. In particular, no politician is ever fully entrenched.*

Holding the politician to a maximal continuation value of w rather than 1 entails a cost of reducing the short-term effort that can be sustained, with the benefit of allowing the voter a fresh start with a new politician rather than retaining an entrenched one. To see why $w \approx 1$ should be sufficient, then, note that the incentive cost is very small in this case, while the benefit is bounded away from zero.

In sum, the model here sheds light on two countervailing forces, generating a cost and a benefit to the voter when public office generates attractive outside offers (i.e., when p and w are high). The cost is that the

[13] This assumption is useful technically, but we do not view it as substantive. The voter and politician could react to commonly observed global events that are unrelated to the productivity shocks that drive the relationship's moral hazard problem.

[14] Various assumptions here are irrelevant. For example, the results are substantively identical if the voter observes the politician's offer, if the sunspot realizes at the end of the period, if a newly elected politician cannot condition on a sunspot, or if a newly elected politician cannot immediately receive the outside opportunity.

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effective discount factor is only $\delta(1-p) < \delta$ whenever the politician's continuation value is below w because she will leave office with a hazard rate p. Intuitively, if the politician is less likely to be beholden to the voter for long, the voter has limited scope for providing dynamic incentives. The benefit is that outside offers give the voter a cost-effective way to reward a successful politician. The voter provides incentives in the current period by meeting failure with future punishment and meeting success with future rewards. But in our baseline model, conferring this reward upon the politician is costly to the voter—having the incumbent shirk in office rather than starting anew with another politician. If, instead, the incumbent takes a lucrative outside offer, the voter outsources an otherwise costly reward to the private sector.

ADDING ADVERSE SELECTION

We now enrich the baseline model by assuming that each politician has a perfectly persistent type $\omega \in \{0,1\}$, which only she knows as of her first term in office. Types are independent across politicians and take the high value of 1 with probability $q \in (0,1)$. A politician's payoffs are as before, but the outcome in each period t produced by a type ω office-holder is $y_t = \omega a_t \theta_t$. Thus, low-type politicians do not produce any value to voters no matter how strongly they are incentivized to do so. We thus refer to type $\omega = 1$ as the good/high type and type $\omega = 0$ as the bad/low type.

As in the previous section, it will be convenient here to assume the voter and politicians have access to a public correlation device. Specifically, we assume a randomization device realizes after the politician's effort decision but before the voter's re-election decision.

A starting observation is that stationary play is again suboptimal if $\delta < c$, as in the baseline model. In particular, two potential sources of nonstationarity exist under a voter-optimal equilibrium—dynamic moral hazard, as in our baseline model, and a learning problem that is itself inherently dynamic.

Another observation is that learning in this model features what the strategic experimentation literature calls "perfectly revealing good news." Under this special structure, the politician, after her first term in office, is either known to be a good type (due to a past success) or is less likely to be the good type than an untried challenger. A feature of the baseline model that carries over to this model is, therefore, that if a politician fails in her first term, she is immediately replaced.

Lemma 1. *In a voter-optimal equilibrium, any politician who fails in her first term is removed from office.*

The lemma implies that the methods that we used to characterize the voter-optimal equilibrium in the baseline setup can be adopted here, as well. Other than in a politician's first term in office, she is known to be good. Therefore, the learning problem need not make an appearance in our recursive analysis.

Proposition 4. *Suppose $\delta < c$. Some $u^s \in [1-\delta, 1]$ exists such that in the voter-optimal equilibrium,*

1. *if period t is her first term in office, a politician of type ω chooses $a_t = \omega\theta_t a_*$, where $a_* := \min\{1, \frac{\delta}{(1-\delta)c}\}$;*
2. *if a politician fails in her first term in office ($y_0 < a_*$) then she is replaced; and if she succeeds ($y_0 \geq a_*$), then she is retained with continuation value u^s. Thus, her continuation value at the start of her career is $u_0 = (1-\delta)(1-\mu c a_*) + \mu\delta u^s$ if she is of high skill and $1-\delta$ if she is of low skill.*
3. *If a politician who has previously produced a success starts a term in office with continuation value $u \in [1-\delta, 1]$, then as in the baseline model, in period t,*
   (a) *the politician chooses $a_t = \theta a_u$ when the state is θ, where $a_u := \min\{1, \frac{1-u}{(1-\delta)c}\}$;*
   (b) *given the outcome y in period t, if $v_u(a_u, y) \geq 1-\delta$, then the politician is retained with probability 1 at continuation value $v_u(a_u, y)$ from the next period. If $v_u(a_u, y) < 1-\delta$, then the politician is retained with probability $v_u(a_u, y)/(1-\delta)$ (determined by the sunspot) at continuation value $1-\delta$.*

The proposition implies that as long as the voter does not know the type of the politician in office, he cycles through politicians until he discovers one that is good. Since bad types always fail, this discovery happens the moment a politician succeeds. After that point, the good politician's career is similar to that in the baseline model, with one important difference: It may be optimal for the voter to start the politician's career at a continuation value that is higher than $1-\delta$—or, equivalently, to provide a newly elected good politician strict incentives to work. The reason for this possibility is that in the model the voter faces a replacement cost. When he boots the incumbent out of office, he gets a flow payoff of zero until he draws a skilled type (and the high shock). Since this may take some time, removing a politician that is known to be good is costly. Based on this cost, the following corollary provides a sufficient condition for the good type's starting continuation value to exceed $1-\delta$, that is, for a newly elected politician to retain some rents.

Corollary 4. *If $\delta < c$ is close enough to c, and μ and q are small enough, then in the voter-optimal equilibrium a newly elected politician has continuation value strictly above $1-\delta$ and, given a good shock, has strict incentives to work.*

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The corollary considers which continuation value for the politician best serves the voter. If that value is low, the politician will likely soon be fired. If it is high, she will likely soon be entrenched. The voter-optimal continuation value for the politician optimally resolves this trade-off. In particular, the corollary provides sufficient conditions for this optimal value u^s to be strictly greater than $c(1-\delta)/\delta$—that is, higher than the continuation value of a politician who, in the previous period, was to be fired for a failure, was made indifferent between maximum and minimum effort, and was successful. These conditions entail q and μ being low. For low values of q, the cost of firing the politician is high, as it is unlikely that the voter will discover a good politician soon after removing the incumbent. For low values of μ, the risk of having to soon reward the politician with full job security is also low, as the likelihood of a success—even if the politician is good—is low. Therefore, at least in the range where the politician exerts full effort, the voter prefers the politician's continuation value to be as high as possible. If δ is close to c, then the range in which the politician exerts full effort will have $c(1-\delta)/\delta$ in its interior. In this case, the voter can provide a good-type politician with strict incentives to work in the first term and still incentivize maximal effort in the next. Therefore, when the cost of firing is high and risk of entrenchment is low, the politician's "one-success" continuation value is lower than the voter would like her to have.

In the standard two-period accountability model, when type and effort are complements and bad types always fail (as we have assumed here), Ashworth et al. (2017) show that no tension exists between selecting good types and sanctioning poor performance. If the voter considers the good type to be better than the bad type even when the good type shirks, then the voter has an incentive to learn the politician's type even if neither type works in the second period. By incentivizing first-period effort from the good type, the voter is simultaneously able to separate the two types.[15]

However, our dynamic setting exhibits a novel trade-off between sanctioning and selection that can arise even though effort and ability are complements. The fact that the voter has a selection problem at present means that he has less latitude to provide incentives in the future. The reason is precisely the replacement cost that we mentioned earlier. If the voter replaces a politician who is known to be good, then it may take a while before he draws another good type. This makes him reluctant to replace a known good type. The more reluctant he is, the higher is the good type's continuation value; and the higher this continuation value, the less the politician tends to work in the future. This intertemporal trade-off between selection and sanctioning is a novel political phenomenon that arises in the multiperiod model but is absent from the two-period model.

[15] Their observation contrasts with the case studied by Fearon (1999) and Duggan and Martinelli (2020) in which effort and type are substitutes. For example, suppose the good type is a commitment type that always works, and the bad type is a strategic type that works only in response to incentives. If the voter succeeds in incentivizing work from the bad type, he makes the bad type look more like the good type, undermining his ability to screen them. Thus, when effort and type are substitutes, there is a trade-off between selecting good types and sanctioning bad performance.

CONCLUSION

The restriction to stationary strategies in infinite-horizon models of political accountability precludes an understanding of how optimal re-election rules shape the careers of politicians. Looking at a model of dynamic moral hazard, we have clarified the link between these re-election rules and the career paths of politicians, showing that optimal re-election rules produce rich career dynamics when the conflict of interest between voters and politicians is large enough. The key observation that comes out of our analysis is that politicians tend to work less in the voter's interest while becoming electorally more secure. An important takeaway from this observation is that one cannot conclude from the fact that a politician becomes entrenched that the voter has not been holding her accountable.

More specifically, we cannot draw broad normative conclusions about whether or not political accountability is at work in a particular context from the fact that politicians have a lifecycle. Even under the normative benchmark of maximal accountability, politicians have evolving careers from their first term in office, when they are most electorally vulnerable, to later terms, when they may have much greater job security. Although it may be tempting to infer an accountability failure from the observation that some politicians appear to have become entrenched, entrenchment alone does not provide sufficient evidence for accountability failure.

ACKNOWLEDGMENTS

We are grateful to Vincent Anesi, Dan Barron, Dan Bernhardt, Peter Buisseret, Steve Callander, Gabriel Carroll, Isa Chavez, Ernesto Dal Bó, Lucas de Lara, John Duggan, Georgy Egorov, Jim Fearon, Dana Foarta, Germán Gieczewski, Edoardo Grillo, Justin Grimmer, Marina Halac, Navin Kartik, Chad Kendall, Cesar Martinelli, Andrea Matozzi, John Matsusaka, Adam Meirowitz, Nicola Persico, Carlo Prato, Ravideep Sethi, Takuo Sugaya, and various seminar audiences for valuable conversations. Apoorva Lal, Ellen Muir, Toby Nowacki, and Sam Wycherley provided excellent research assistance. A previous draft of this paper was circulated with the title "Optimal Political Career Dynamics."

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Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Acharya, Avidit, Elliot Lipnowski, and João Ramos. 2025. "Political accountability under moral hazard." *American Journal of Political Science* 69: 641–52. https://doi.org/10.1111/ajps.12860

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